

PricePal Running App: Exploring New Ways of Motivating Physical Activity

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Abstract

PricePal incentivizes physical activity through reward-based challenges sponsored by brands, which allows users to get discounts when completing fitness milestones. This approach merges business goals with public health benefits by promoting a healthier, more active lifestyle, while fostering community engagement and brand interaction. Key elements of the app's development include comprehensive competitor and stakeholder analyses, iterative prototype design, and user testing to refine the app's functionality and user experience. Initial financial projections suggest long-term viability, highlighting a positive net present value and robust revenue potential. This report also explores the app's technical framework, covering potential integration with third-party APIs and native tracking solutions. Our findings underscore PricePal's capacity to drive meaningful engagement in physical activity, leveraging motivational theories like the IKEA effect and financial incentives to encourage sustained user involvement.

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1. Introduction

Physical activity is essential for society since it plays a critical role in promoting both individual and public health. Regular physical activity reduces the risk of chronic diseases such as heart disease, stroke, diabetes, and certain cancers, which are major contributors to global mortality. Beyond physical health, it supports mental well-being by lowering rates of depression and anxiety, improving mood, and enhancing cognitive function. For society at large, increased physical activity can lead to reduced healthcare costs, improved productivity, and a higher quality of life. (World Health Organization, 2024)

To help reduce the global burden of non-communicable diseases and improve mental and physical health across the lifespan the World Health Organization (WHO) provides recommendations on physical activity across all age groups. For adults, WHO recommends at least 150 to 300 minutes of moderate-intensity aerobic activity, for instance brisk walking, or 75 to 150 minutes of vigorous-intensity activity, like jogging or running, per week. Additionally, muscle-strengthening activities should be performed on two or more days weekly. The WHO also highlights the need for reducing sedentary behaviour, recommending that all individuals should minimise prolonged periods of inactivity, which can negatively impact health even in physically active individuals. (World Health Organization, 2020)

The rising cost of living can significantly impact physical activity levels, particularly in lower-income groups. As household expenses increase, many individuals are forced to reduce or cancel paid activities such as gym memberships and organised sports. This shift is more pronounced in disadvantaged communities, where basic needs like food and housing take priority. While some people may turn to more affordable or free alternatives like walking or home workouts, the overall reduction in access to structured physical activities worsen existing inequalities in participation and health outcomes. (Sport England, 2024)

To combat these societal issues of sedentary behaviour, lack of physical activity and high costs of living, we present PricePal. The app aims to encourage users to get active by completing challenges set by companies in exchange for exclusive discount codes. By rewarding physical activity with tangible benefits, PricePal makes staying active more affordable, accessible, and motivating for all. This innovative approach not only encourages healthier lifestyles but also builds a community of engaged, goal-oriented individuals, helping to close the gap in health imbalances and inspire positive change in everyday life. PricePal is more than an app; it is a tool for a healthier and more active everyday life.

2. Approach and delimitations

2.1 Brainstorming

We and five other members of the course were pitching our ideas to each other after a lecture. One standout concept was an app enabling friends to collectively contribute money into a pot, effectively allowing them to bet on their own achievements. The premise was that participants would share the winnings if they succeeded in their goals, while those who did not would forfeit their contributions. For example, a group of friends might use this app to motivate one another to complete a marathon. As our discussions progressed, the idea evolved into an app that would allow companies to host events where users could participate with their friends and place money in a collective pot. During

this meeting, we also introduced the concept of participants earning rewards for their achievements. Finally, our group decided to focus on this reward-based running initiative.

2.2 Initial Viability Analysis

From a business perspective, we envisioned that companies could purchase space on our platform to host challenges, thereby gaining visibility and attracting potential customers who might be incentivized by discounts for participation. We outlined key components necessary for developing a Minimum Viable Product (MVP). These core features included:

- the ability for individual users to join challenges
- activity logging
- issuance of rewards upon challenge completion

Additionally, we began analysing the competitive landscape, identifying key competitors like Stepler, RunKeeper, Strava, Mecenat, Studentkortet, and Fantasy Walk. This demonstrated that the market is viable and that our concept effectively addressed an existing market gap.

To ensure different customer needs are met, we want to include a broad range of advertisers. This would not only provide more varied rewards, but also attract a more diverse audience, making the app more inclusive and usable.

3. Previous research

Research by Mitchell et al. (2020) published in the British Journal of Sports Medicine highlights the effectiveness of financial incentives in increasing physical activity levels, particularly among adults. This study demonstrates that monetary rewards, when paired with specific activity goals, drive substantial improvements in exercise engagement. For example, offering users discounts on training gear for achieving milestones, such as running 10 kilometres, can stimulate a positive reinforcement cycle. In this model, rewards act as immediate positive reinforcement, strengthening the association between physical activity and tangible benefits, ultimately encouraging individuals to repeat the behaviour.

The concept of “enclothed cognition” offers additional insight into how rewards tied to athletic apparel can motivate people to be active. Michele Ross (2023) discusses how wearing attire specifically associated with fitness impacts psychological processes, which in turn can influence behaviour. When individuals wear clothes designed for physical activity, they are more likely to view themselves as active, which increases their motivation to exercise.

Furthermore, research from Harvard Health reveals that when individuals receive a reward tied to a health-related goal, their intrinsic motivation often increases (Solan, 2024). The key to sustainable engagement with fitness is transforming initial motivation into long-term habits. Financial incentives are excellent for jumpstarting exercise routines, but our goal extends beyond one-time participation to fostering lasting commitment. Research emphasises that incentives can be especially effective in promoting habit formation when paired with achievable goals and immediate rewards. As the Harvard Health study suggests, immediate rewards are more likely than long-term ones to encourage repeated behaviour (Solan, 2024).

3.1 How PricePal uses these findings

By implementing fitness challenges that reward users with discounts on popular brands like Nike, our app leverages this evidence-based approach to motivation. Financial incentives become more than a one-time motivator; they serve as a crucial component in the process of habit formation, helping users develop consistent exercise routines. This aligns with Mitchell et al. (2020) that shows how small, consistent incentives can play a pivotal role in overcoming the initial barriers many face in establishing a new fitness habit. Furthermore, by setting achievable challenges and providing immediate discounts upon completion, our app helps users feel a sense of accomplishment and satisfaction that keeps them engaged in their fitness journey.

Our app's discount model taps into the psychological effect of enclothed cognition, offering users rewards that make quality athletic wear more accessible. This approach not only motivates users through financial savings but also encourages them to perceive themselves as fitness-focused individuals, promoting a positive identity shift.

4. Business

Our business process was divided into four main steps: a competitor analysis, a stakeholder analysis, a market survey, and a financial analysis. All these are described more in detail in the subsections below.

4.1 Competitor Analysis

A competitor analysis was an important step in finding our niche and what sets our idea apart from similar services. We could also observe which solutions could inspire us and what aspects we should change to achieve our goals. The analysis also led to a viability analysis in which we were able to identify that each element of our idea has been successfully done by other companies, but not with the combination we had in mind. This indicated that the idea should be viable from both a technological and business perspective and should be able to attract both users and advertisers. After completing our competitor analysis and viability analysis the natural follow-up was to articulate the value we create and how that differs from our competitors.

We have not found any platform that combines the ideas of advertising, as seen in Mecenat and Stepler, with community, tracking, and competition, as seen in Strava, Runkeeper, and Facebook groups. This is where our niche lies. A potential challenge for attracting users however is that apps like Stepler generate rewards for their users passively, requiring little to no effort, while our app demands commitment through challenge completion which requires more active engagement.

The advantage we believe our model has over Mecenat or Stepler is that challenges should generate more engagement with the brand than simply showing an advertisement and providing a discount code. If a user actively participates in a specific brand's challenge and runs a mile to complete it, this should positively influence their perception of the brand far more than just receiving a direct discount code. Additionally, effects related to the IKEA effect could come into play, where users value the product more because they have invested effort in achieving it. This is illustrated in a study by Norton, Mochon, and Ariely (2012) published in the *Journal of Consumer Psychology*, which demonstrates that this effect persists regardless of an individual's usual preference for putting effort into achieving results.

The IKEA effect should result in us generating a higher value per engagement compared to traditional ads on other platforms, where no effort is required from the user. Consequently, we should be able to charge a higher price. However, the requirement for user effort likely means we won't be able to reach the same volumes. The downside of the IKEA effect is that users tend to devalue things if they are unable to participate in the creation process or if their efforts do not lead to success (Norton, Mochon, & Ariely, 2012). This could mean that users who fail to complete a challenge might actually value the product less, potentially damaging the brand. To mitigate this problem challenges will have continuous rewards, for example you get 5% off if you run 1 km, 10% off if you run 5 km and 20% off if you run 10 km. This would make it much easier for the user to at least reach a goal and thereby minimise the risk that they feel like they've failed and then project these negative emotions onto the brand. What will be best for the brands and users will require further investigation.

We have chosen to signal different values compared to our competitors. While Mecenat may be more associated with budget and low-cost offerings, we wanted to position ourselves more in line with health, wellness, and fitness. This should allow companies to influence their brand image differently by advertising with us, as opposed to advertising with our competitors.

4.2 Stakeholder Analysis

In order to get a better understanding of how the project would impact and be impacted by different actors, we conducted a stakeholder analysis assuming the product was launched and that the company had recruited employees to development and management. We used the Mendelow's Matrix (Professional Academy, 2021) in which relevant stakeholders are put on a matrix based on their interest in and power to impact the project (see Figure 1).

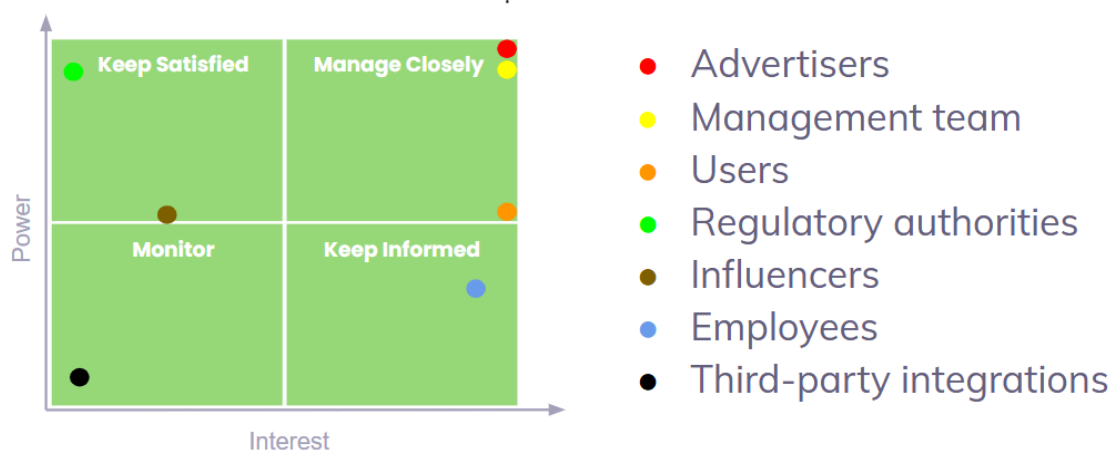


Figure 1: Stakeholder analysis matrix.

The analysis found that the most important actors are the advertisers, the management team, and the users. These actors should be managed closely, in line with the model, in order to use their power in ways which benefit the service. Employers are put in the “Keep informed” quadrant, as they work with the project and thus should always be kept up to date with what is going on. The Influencer actor refers to individuals or organisations which market PricePal; the name was chosen as to not be confused with advertisers. Together with regulatory authorities, they have a clear influence over the regulation and reputation of the service. Lastly, third-party integrators are placed in the bottom left “monitor” quadrant. The third-party integrators have a bit more power as they can impact what kind

of data we can access from their services. However, since we aim at further developing our own tracking system these third-party integrators do not affect the core functionality of the app.

4.3 Market Survey

A market survey (see [Appendix A](#)) was conducted in order to get a better understanding of how our idea would be perceived by our advertisers as well as gain insights to further refine our idea. The survey aimed to assess the preferences of potential advertisers regarding different promotional models. Participants were first asked to rank three types of companies based on their interest in working with them: those providing discount codes to specific groups (like students or retirees), those offering traditional advertising on a platform, and those giving discount codes that customers earn by completing a challenge (such as running a set distance). The survey also inquired whether advertisers found it more valuable to offer discounts that customers must "earn" through activity challenges, as opposed to discounts given directly. Respondents could choose whether they found this model more valuable, equally valuable, or less valuable than straightforward discount codes. Respondents were asked to provide a brief explanation for their ranking choices on both questions.

The survey was chosen to be concise and only focus on whether the mechanism of needing to complete a challenge was seen as valuable or not by the advertisers as we argued that this would be the key component in making our idea successful or not. If the advertisers do not see the benefit then there is no reason for them to use our platform compared to our competitors where they may achieve a larger volume. It would then be necessary to convince the advertisers that our form of marketing would provide more value per sale compared to our competitors, which could present a significant hurdle for our expansion.

The survey was also chosen to use comparative questions rather than for example asking users to rank the idea on a scale of 1-10. This is to avoid the phenomena of people generally thinking any idea one proposes is good. The rationale is that if we instead "force" respondents to compare what value we bring to what our competitors bring we will gain much more insights on how good our idea actually is as well as what its strengths and weaknesses are in the landscape where it will then be deployed.

The survey was then sent out to 8 people working in marketing, from whom we received 3 answers (see [Appendix B](#)). For the first question, two companies ranked the option where companies provide discount codes if a challenge has been completed, as their top choice, while one company ranked the option of providing discount codes limited to a specific group, as the highest. Responses included statements like *"Simpler – don't overcomplicate things"*, *"There's a greater chance customers will use the discount code if they've done something to earn it, but we also want to reach as many people as possible"*, and *"This would work well for our outdoor-focused target group."* For the next question, all companies agreed that it is more valuable for customers to earn discounts by completing tasks. Motivations for this preference included responses such as *"Engagement is valuable"*, *"Engaged customers are more willing to buy"*, and *"I believe they would be more loyal"*. These answers indicate that companies are interested in reaching a dedicated and engaged customer base.

The results suggest that companies prefer providing discount codes as a reward for customers completing specific challenges. This approach is seen as more likely to engage a broad audience, especially in markets with active or outdoor-focused consumers. Companies value task-based discounts for fostering customer engagement, which they believe leads to higher purchase intent and loyalty. Overall, the preference highlights a strategic focus on building a dedicated and involved customer base through interactive incentives.

4.4 Financial Analysis

In our financial analysis we have examined the net present value (NPV) and long-term viability of the app. The values in the financial analysis are hypothetical, however based on well formatted assumptions. The cash flow projection for the company, as outlined in Table 1 below, shows a steady progression from negative to positive cash flows over a seven-year period, reflecting the company's business growth and profitability. By examining the company's financial performance year by year, we can gain deeper insights into how different revenue and cost drivers impact its financial sustainability and potential for long-term success.

Cash Flow Analysis

Year	0	1	2	3	4	5	6	7
Inkomst Challenges Sek	- kr	- kr	1 920 000 kr	1 920 000 kr	3 000 000 kr	3 000 000 kr	3 000 000 kr	4 320 000 kr
Inkomst Kompensation Sek	- kr	1 000 000 kr	2 250 000 kr	4 218 750 kr	6 328 125 kr	9 492 188 kr	17 085 938 kr	25 628 906 kr
Utgifter Sek	1 000 000 kr	2 000 000 kr	3 000 000 kr	4 000 000 kr	5 000 000 kr	6 000 000 kr	7 000 000 kr	8 000 000 kr
Cash Flow/Year Sek	- 1 000 000 kr	- 1 000 000 kr	1 170 000 kr	2 138 750 kr	4 328 125 kr	6 492 188 kr	13 085 938 kr	21 948 906 kr
Total Cash Flow Sek	- 1 000 000 kr	- 2 000 000 kr	- 830 000 kr	1 308 750 kr	5 636 875 kr	12 129 063 kr	25 215 000 kr	47 163 906 kr
NPV	9 857 719 kr							

Table 1: Cash flow analysis from year 0 to year 7.

In Year 0, we incur an initial cost of 1,000,000 SEK, for marketing and app development, but generate no revenue since the app has not yet launched, resulting in a negative cash flow of -1,000,000 SEK. This phase is critical for setting up our presence in the market and creating awareness of the app among potential users.

In Year 1, we'll begin generating revenue through kickback from customer purchases, with 50,000 purchases made at an average price of 20 SEK per purchase, resulting in total revenue from purchases of 1,000,000 SEK. However, with increased expenses of 2,000,000 SEK, the company still experiences a negative cash flow of -1,000,000 SEK. Although this year shows financial strain, the introduction of purchases indicates the beginning of customer engagement with the app. The continued marketing investment is necessary to build momentum and attract a larger user base in future years. The 50 000 purchases is estimated from our expectation to get around 4000 users who make approximately 1 purchase each month.

In Year 2, the company starts to diversify its revenue streams. In addition to purchase revenue, which increased to 2,250,000 SEK from purchases increasing by 50% to 75 000, now we will start charging for the challenges, each costing 8,000 SEK. This generates an additional 1,920,000 SEK in revenue from challenges, with 20 challenges a month. With total income of 4,170,000 SEK and expenses of 3,000,000 SEK, gives a positive cash flow of 1,170,000 SEK for that year.

In Year 3, the company further expands its revenue sources. The number of purchases again are expected to rise by 50% to 112 500, with the average price per purchase increasing to 25 SEK, yielding 4,218,750 SEK in purchase revenue. At the same time, we continue to run challenges, maintaining 20 challenges per month at 8,000 SEK each, adding 1,920,000 SEK in challenge income. Total revenue for the year reaches 6,138,750 SEK, and after covering expenses of 4,000,000 SEK, the company achieves a substantial cash flow of 2,138,750 SEK and a positive total cash flow of 1,308,750.

In Year 4, the business shows continued robust performance. Purchase revenue grows to 6,328,125 SEK, from a 50% increase in purchases. This continues to increase by 50% per year which gives a revenue of 9,492,188 SEK for year 5. We will increase the number of challenges to 25 per month for year 4 and 5, with each challenge now priced at 10,000 SEK, bringing in 3,000,000 SEK from challenges.

By Year 6, the company further capitalises on its market position. The average price per purchase increases to 30 SEK, with almost 570 000 purchases in the year generating 17,085,938 kr SEK in revenue. Challenge income remains consistent, adding 3,000,000 SEK for the year. Total income is 20,085,938 SEK, with expenses rising to 7,000,000 SEK, leading to a cash flow of 13,085,938 SEK.

In Year 7, purchase revenue reaches 25,628,906 SEK from around 854,000 purchases, each priced at 30 SEK. Additionally, we will increase the number of challenges to 30 per month, charging 12,000 SEK per challenge, contributing an additional 4,320,000 SEK in challenge income. Total revenue for the year is 29,948,906 SEK, while expenses expect to increase to 8,000,000 SEK, leading to a record cash flow of 21,948,906 SEK and total cash flow of 47,163,906.

The Net Present Value (NPV) of 9,857,719 SEK indicates that the company's future cash flows, when discounted for risk and time, far exceed its initial and ongoing investments. A positive NPV is a clear sign of financial viability and suggests that the company is creating significant value for its investors. This figure underscores the long-term profitability potential of the business, making it a compelling investment opportunity.

4.5 Business Analysis Summary

Our competitor analysis focused on identifying our niche within the current market by evaluating competitors like Mecenat, Stepler, Strava, and Runkeeper. This analysis confirmed that our unique combination of advertising, community, tracking, and competition is viable, and likely to attract both users and advertisers. Additionally, we identified the potential for increased brand engagement through "effort-based" rewards, leveraging concepts like the IKEA effect, which increases perceived value for products earned through personal effort.

The stakeholder analysis used Mendelow's Matrix to map out key stakeholders, identifying advertisers, management, and users as primary actors to manage closely due to their high interest and influence on the project's success. Clearly understanding and prioritising stakeholder interests, can ensure effective alignment and support for the project's growth.

Our market survey was conducted with marketing professionals to assess the appeal of challenge-based discounts versus traditional discount methods. The results highlighted that advertisers value committed customers who earn discounts through challenges, viewing this as more engaging than standard discount codes. This suggests that our app's challenge-based approach could provide advertisers with high-quality customer engagement and loyalty.

Lastly, our hypothetical financial analysis examined the business's long-term viability, projecting a positive cash flow trajectory over seven years. Initial high marketing expenses lead to negative cash flow, but increased purchases and paid challenges drive revenue growth, ultimately achieving a NPV of 9,857,719 SEK. This positive NPV signals strong financial sustainability and an attractive investment opportunity, supported by consistent revenue growth and expanding user engagement.

5. Design

5.1 Ideation

Our team's first step in defining the app's user experience involved a collaborative brainstorming session, where each member brought and presented a low-fidelity, paper sketch of the user flow, see

Figure 2. These sketches, based on the previously agreed-upon MVP features, served as visual interpretations of the app's core functions. During the meeting, we focused on clearly communicating our individual ideas and discussing different design approaches, enabling a shared exploration of each member's vision for the app. This method was chosen as it is a cheap and very fast way to generate, discuss and improve on a large number of ideas (Dam & Siang, 2024).

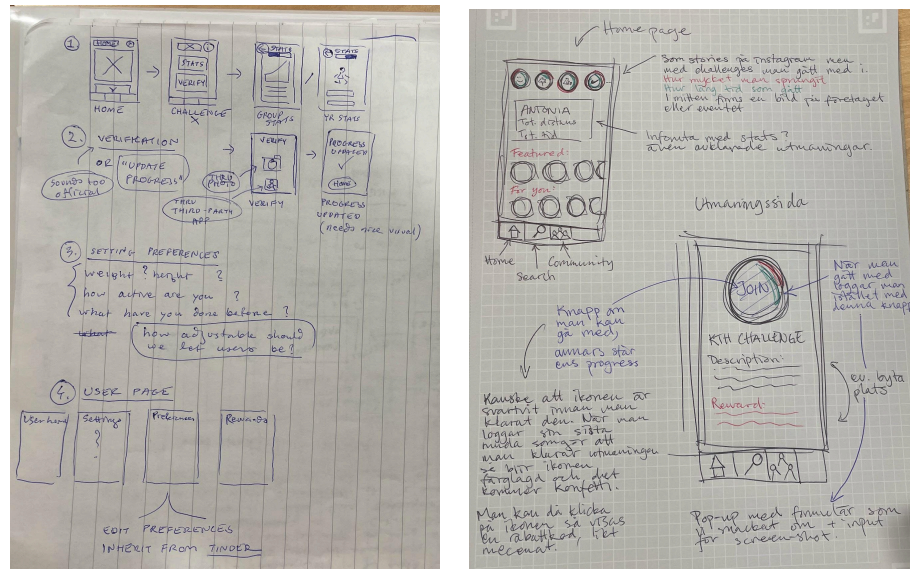


Figure 2: Selection of paper sketches used in the collaborative brainstorming session

Using this process, we combined elements from each sketch into a new, cohesive paper prototype. The synthesis of multiple viewpoints led to a stronger initial design that more comprehensively addressed user needs and interactions. As a result, this first brainstorming session laid a solid foundation for developing our prototype, and provided clarity for the next phases of development, setting us up to refine the app's look and feel in a more focused, iterative process.

5.2 Low-Fidelity Prototypes

To bring our initial design concept to life, we developed a series of lower-fidelity prototypes using Figma as seen in Figure 3 and Figure 4. This tool allowed us to quickly create and iterate on the design, moving through successive stages of refinement as we gradually enhanced the level of detail. Throughout this process, we leveraged continuous feedback from our team, ensuring each iteration was aligned with our collective vision and improving the design through frequent, collaborative design reviews.

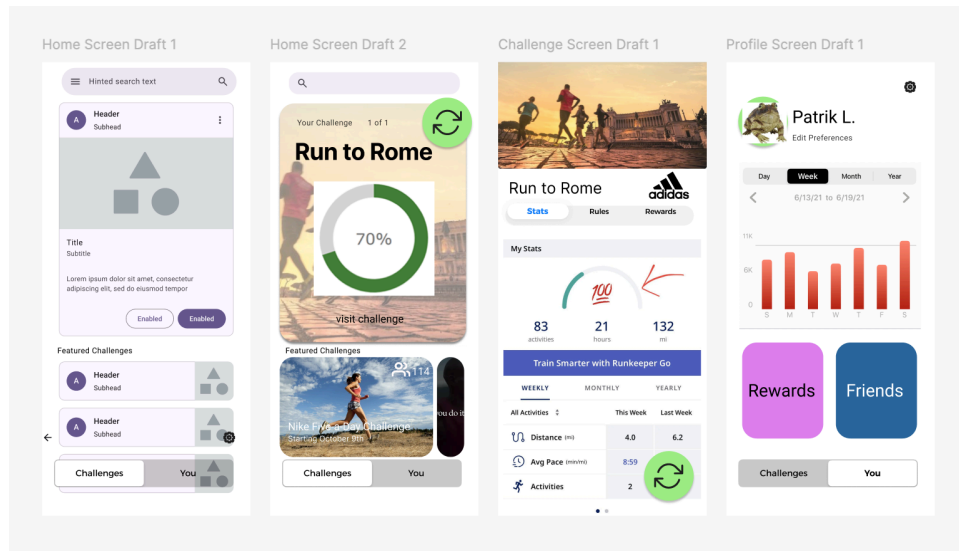


Figure 3: The first iteration of the prototype in Figma

We chose this internal iterative approach because it allowed us to rapidly develop the design. While user integration would probably have been insightful at this stage, the time that scheduling meetings, having interviews and analysing results requires were not considered to be a high enough priority. To make up for not conducting our own user testing at this point, we analysed the design of established platforms such as Runkeeper and other popular apps.

Drawing insights from established apps also aligns with the fourth heuristic of user interface design, Consistency and Standards, as it makes our app more consistent with users expectations (Nielsen, 2024). This strategy provided us with a valuable shortcut, enabling us to build on proven user interactions and apply them to our app in a way that suited our unique goals.

In the second iteration of our prototype as seen in Figure 4 below, we focused more on enhancing the app's interaction flow and user navigation. This version was developed through the same iterative process as before, involving constant feedback and design reviews within the team. We used the basic layout from the first iteration as a foundation but made targeted adjustments to address issues identified in our previous review sessions.

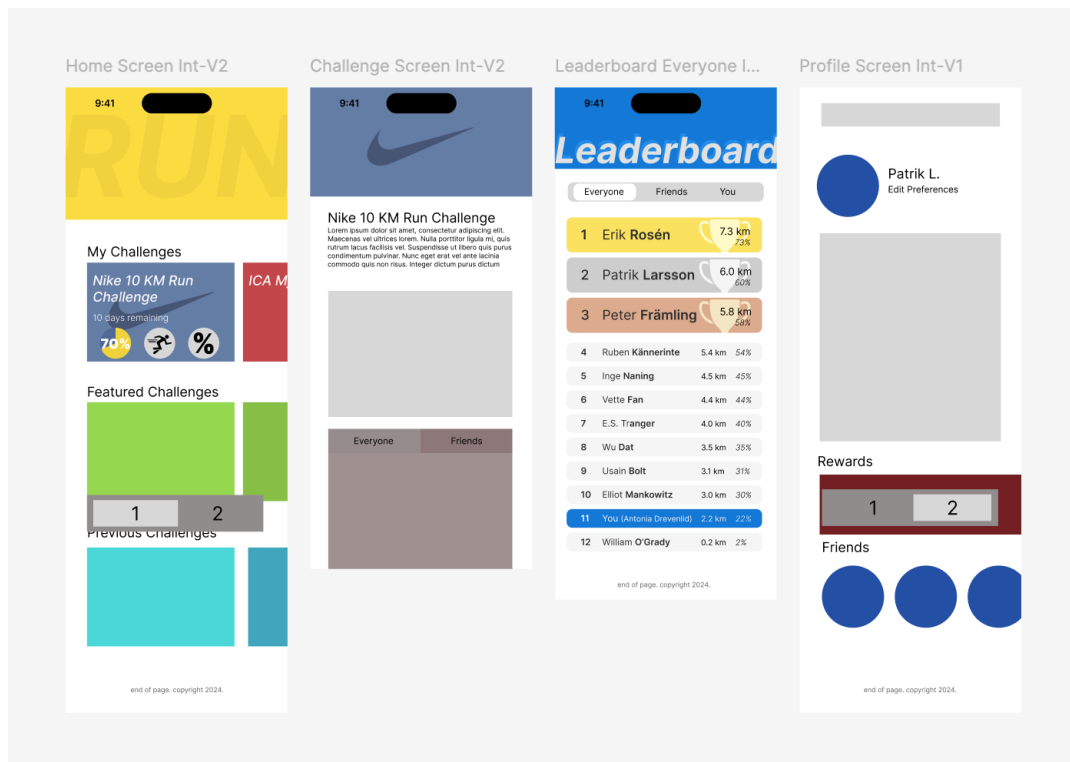


Figure 4: The second semi-interactive iteration of the prototype in Figma.

Key modifications included simplifying the overall layout to display only essential information, reducing the potential of overwhelming the users. We also introduced familiar icons in place of some text to streamline the interface and make key elements more recognizable at a glance.

The primary goal of this iteration was to minimise the number of actions needed for users to complete tasks while maintaining clarity and coherence in the design. Achieving this balance proved challenging, and we quickly realised the importance of validating this flow with actual users. Additionally, while this prototype was effective for our team to communicate and understand the app's intended functionality, we anticipated that real users, without our shared vision, might struggle to navigate this version effectively.

With this initial user flow established in Figma and the apparent need for user validation in mind, we began preparing for a more detailed interface design. This next step would better support usability testing with users, giving us a clearer picture of how intuitive and efficient the app's navigation truly was.

5.3 High-Fidelity Prototype

By this stage, we came to the conclusion that the basic design, navigation and flow developed through the internal design reviews would not generate sufficient insights to move the prototype further without actual user testing. But to avoid the user testing yielding trivial insights we concluded that a high-fidelity prototype was crucial. If we had conducted user testing on the earlier lower fidelity versions, we anticipated that feedback would likely focus on surface-level improvements, such as making the app's design feel more in line with other popular apps on the market today or enhancing its overall professional look and feel. These changes, while valuable, were ones we planned to address either way. By testing a high-fidelity prototype we could ensure that users would provide more actionable feedback on specific design details and usability aspects, rather than on general aesthetics.

Therefore, the high-fidelity prototypes were made with greater attention to detail and with contemporary design practices in mind. This shift allowed us to validate the app's design more in-depth, using a user-centred approach to enhance the user experience based on specific feedback from this stage onwards. This approach maximised the value of our testing resources and allowed us to focus on refining aspects of the design that genuinely impacted the user experience.

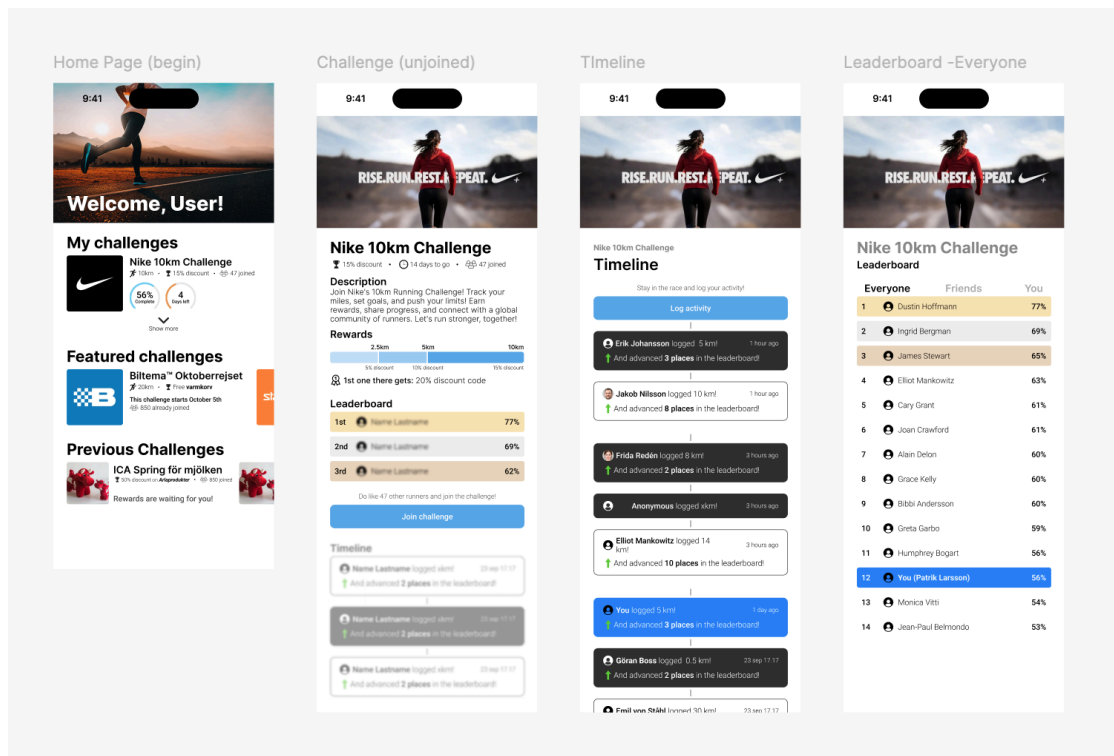


Figure 5: A selection of screens from the first high-fidelity prototype in Figma

5.4 User Testing

5.4.1 Pilot Study

A pilot user study was conducted for two main purposes. First, it provided an opportunity to refine our moderation techniques and the test configuration in preparation for the main user testing sessions in a low-stakes setting. Second, the pilot study served to validate whether the design decisions effectively communicated the intended business concept to users. The prototype used in this pilot study was the prototype shown in Figure 5.

The format of the study:

A. User tasks:

1. Read onboarding & start app
2. Join a new challenge
3. Log activity
4. Check Leaderboard
5. Check your stats
6. Approve a friend request
7. Use a reward

B. Pitch the App: discuss its purpose, its goals, and its intended target user

Participants were asked to complete tasks using only the information available within the prototype, with moderators giving minimal guidance, limited to statements such as “The current task is to join a new challenge” or “The next task is to check your stats”. We encouraged participants to use the *think-aloud method*, enabling us to capture their real-time thoughts and impressions as they navigated the prototype. After completion of all tasks, the participants were asked to pitch the app’s purpose as they understood it back to us.

Three fellow students from KTH were recruited for this pilot study. All three participants successfully identified the problem our app aimed to address and recognized how it could benefit them and others. In terms of the layout and design, one participant described the app as having an “intuitive layout and being easy to navigate”, similarly another participant stated that the “information on the screen is easy to understand”. This initial feedback validated our design choices, proving it to be relevant and fitting to the context.

As anticipated however, the users encountered multiple challenges completing tasks and misunderstood certain interactive features. One participant felt that the app was “overwhelming with too much text” in the challenge view while another found it hard to navigate backwards in the app stating “I usually use the back-button”, a button which did not exist in this design.

Using only three participants in this pilot study, we successfully gathered valuable insights on how to make the experience better, but crucially, the users clearly grasped the app’s overall vision and purpose which was most important in this stage.

5.4.2 User Study, Round 1

Changes made from the pilot study to the first real user testing session included adding a semi-structured interview with seven questions after the tasks had been completed, replacing the reversed pitch. Smaller design changes were also implemented in the prototype based on the insights gathered, such as “back” buttons enabling simpler backwards navigation and more coherent typography styles, see Figure 6.

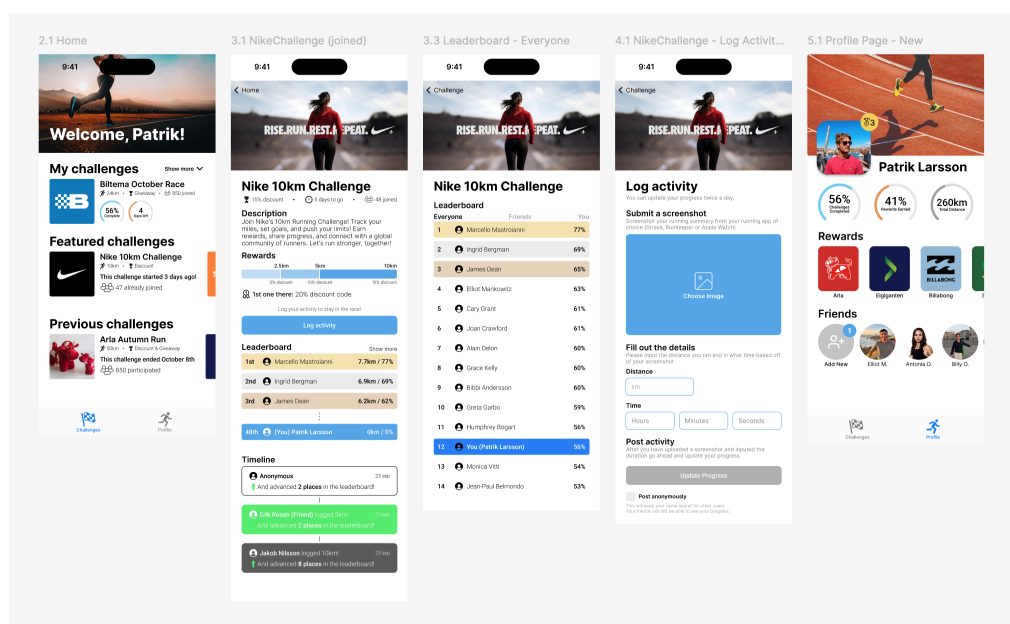


Figure 6: A selection of screens used in first round user testing

The questions for the semi-structured interviews were:

B1. *Have you used similar apps before? How does this app compare to them?*

B2. *Can you describe how to log activity anonymously?*

B3. *Describe how you experienced the layout? Were the elements on the pages what you expected or were you surprised?*

B4. *Describe the flow! Is there a redundant page or a page missing?*

B5. *How did it feel to use the app?*

B6. *Any other comments?*

The main objective of the first round of user testing was to identify problems with layout, design elements, flow and functionality in the prototype. A secondary goal was to understand how users positioned our app within the landscape of similar products, aiming to further validate if the users understood the vision of the app. The problems identified by the participants in the first round would be considered and improved for the second round participants to validate.

This round of testing involved five new participants, different from those in the pilot study. This number was chosen according to usability research by the Nielsen Norman Group, that states testing with just five participants often reveals the majority of usability issues (Nielsen, 2000). This approach leverages the principle that, after a few participants uncover the most significant usability problems, additional participants tend to repeat issues already identified.

One moderator guided the participants through the tasks, and conducted a follow-up semi-structured interview after the tasks were completed. The second moderator focused on taking detailed notes. These notes captured a blend of observed behaviours, such as errors made, time taken for specific tasks, or actions that contradicted the design, as well as participants' spontaneous thoughts about the app. Often, participants would answer our interview questions naturally through their reactions, so it was crucial to record their thoughts as they arose, rather than relying solely on our preset questions.

One bias to consider is that participants' opinions may have been shaped by our specific phrasing of questions. Sometimes, that information would present itself naturally observing the user while doing the tasks. Other times, reflecting back on the app during the interview stage provided more substantive information, when we allowed ourselves to have a dialogue with the participants, asking follow-up questions and the users giving suggestions for changes or expectations. Collecting data in these two distinct ways gave us a very clear indication where the prototype could be improved and how the changes would affect the users.

The responses from the study identified many valuable sources of improvement and insights for the prototype. After the study, each comment was transcribed to a post-it note and placed on a FigJam board for theming and subsequent analysis - see Figure 7. Each colour corresponded to a participant, and each comment was grouped into categories of where in the prototype the error/insight occurred and which combination of Norman's six principles of interaction design, Visibility, Feedback, Constraints, Mapping, Consistency & Affordance, they relate to (Norman, 2002).

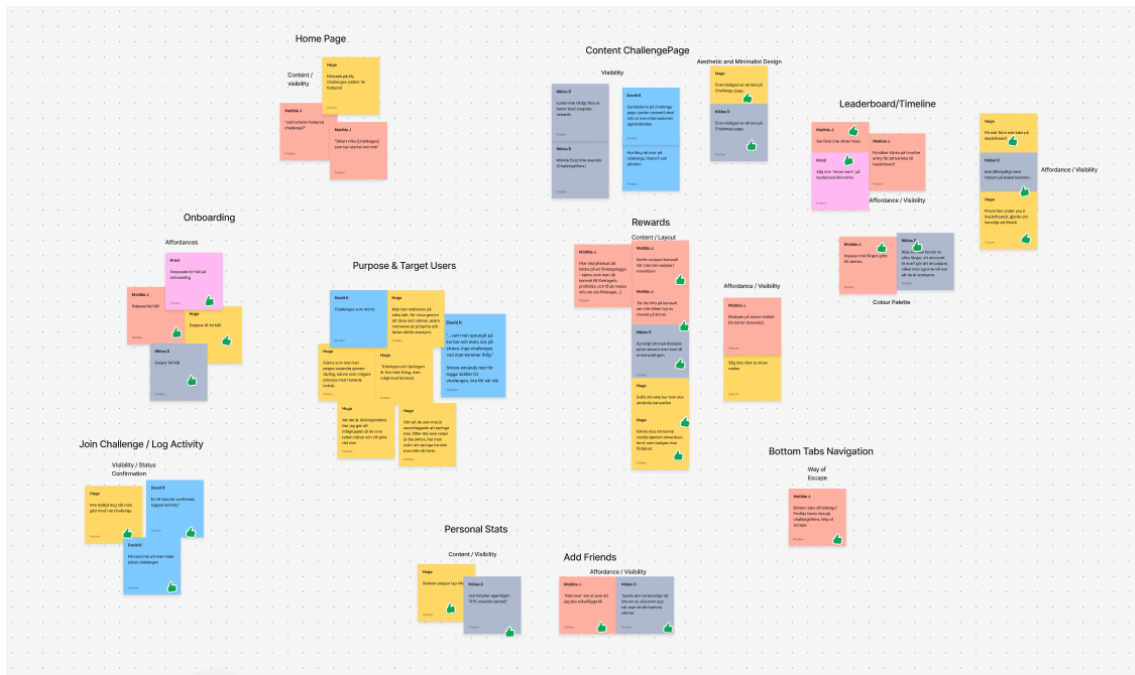


Figure 7: Affinity Mapping of User Testing Responses

Results showed most insights concerned the layout of the Challenge view, where issues such as not receiving sufficient feedback on successful activity logging was commonly stated by the participants. The rewards page functionality was also often misunderstood and not clear enough for the users to intuitively know how to use it. Some of the prototype's affordances were in many cases also not visible enough, the “show more” button next to your joined challenges, which would expand a modal to show all of your currently joined challenges was commonly referenced by participants. Another example of lacking affordance was observed in many participants when navigating between different states of the timeline & leaderboard elements, calling for more intuitive and visible design.

The challenge view layout was identified as the view with most usability issues as it contained too much information at the same time, making the users unsure of what elements were interactive and much of the information and functionality available was “hidden in plain sight” as a result. For instance, the flow from logging an activity to it appearing on the timeline confused some participants, and the distinction between the timeline and leaderboard made the task of *Check Leaderboard* after *Log Activity* in most cases the slowest task to complete.

5.4.3 User Study, Round 2

Based on insights gathered from the first round of user testing, we made several targeted adjustments to improve the prototype for the second round of user testing. For starters, we adapted the app to better accommodate specific requested user flows. For example, if a user opens the app only to redeem a reward, they should be able to access it directly, without navigating through the Profile page. To facilitate these flows, we expanded the bottom navigation bar from two icons (Home and Profile) to four, adding Run Tracker and Rewards. Key updates also included adding tabbed elements to the layout, allowing users to switch between different types of information without overwhelming them. By limiting the amount of information displayed at any one time, we aimed to help users fully absorb each section's content, while making it clear that additional information was available through the tabs. The tabs also pronounced the movement between different states of the prototype, making the visibility of system status more clear. The new design implementations are shown below in Figure 8.

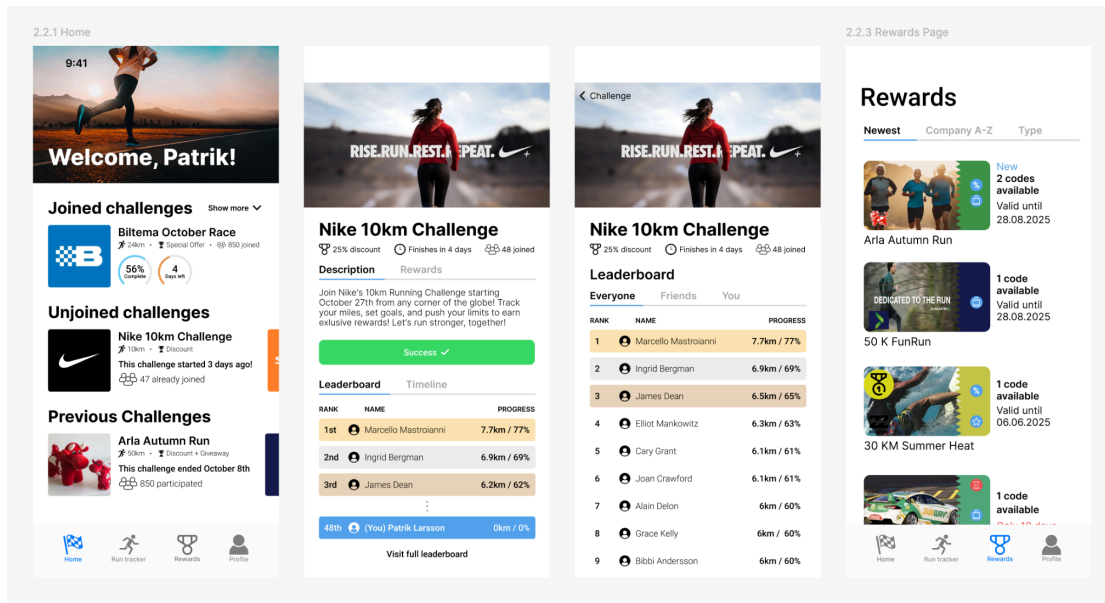


Figure 8: A selection of screens showcasing the result of the first round of user testing

The second round of user testing followed the same format as the first, using the same tasks and interview questions. For this user test the prototype shown in Figure 8 was used. This round included four new participants, along with one participant from the first round. The returning participant provided an additional layer of validation, as her experience with both versions of the interface helped assess the effectiveness of the improvements we had implemented.

Since this round was conducted near the end of the development cycle, the primary goal was to evaluate whether the adjustments had successfully addressed the key issues identified in the initial user testing. The results were encouraging: feedback was largely positive, with participants expressing satisfaction regarding the app's design, layout, clarity, and overall cohesiveness. Most importantly, there were minimal observations related to the issues uncovered in the first round, suggesting that the recent changes had effectively resolved many of the primary concerns.

However, the user-centred design process continually reveals new insights, and this round of testing identified additional areas for potential refinement. One participant noted confusion regarding the differing affordances for viewing additional challenges on the home page. While one type of challenge used a “show more” button, the other allowed users to scroll to view more options. This difference was originally intended to emphasise the distinction between the two types of challenges and to add a dynamic feel to the interface. However, for some users, this approach led to confusion rather than clarity.

5.5 Final prototype

The final prototype (available to try - see [Appendix C](#)) is a high-fidelity, fully interactive experience that showcases our business vision and includes improvements made to fix the most important usability issues discovered in the user studies, see Figure 9.

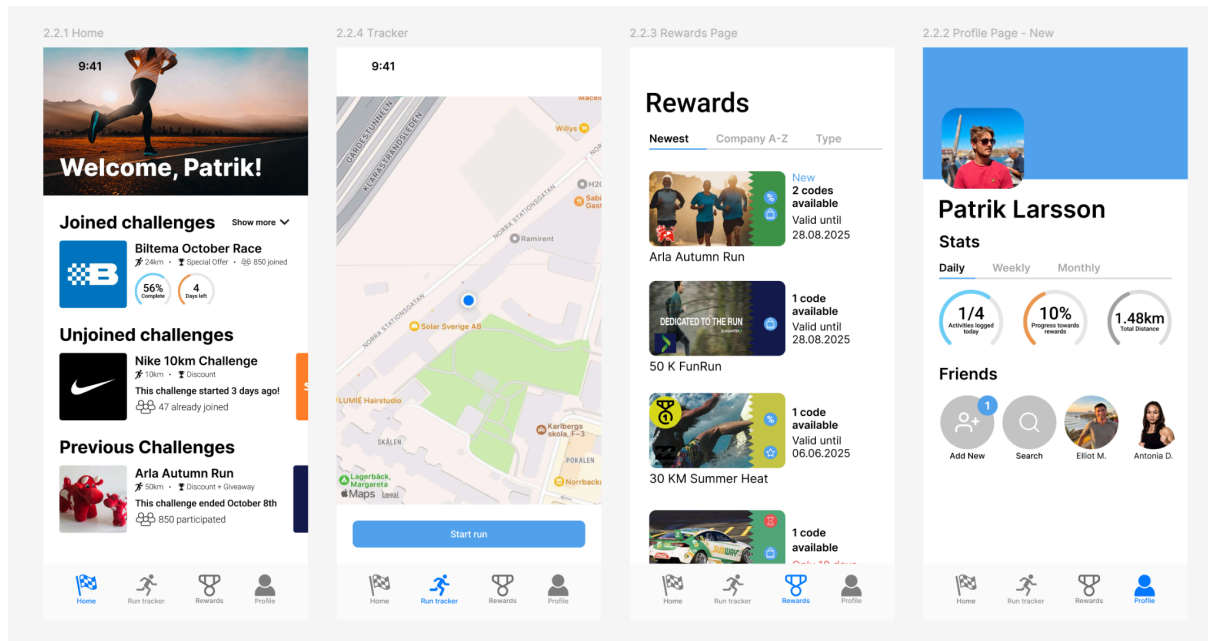


Figure 9: The four main screens of the final prototype

Following the positive feedback from the second round of user testing, we decided not to implement any major design changes to the prototype. The feedback indicated that our recent adjustments to the navigation, layout, and overall user experience had effectively addressed the primary usability concerns raised in earlier testing.

At this stage, a key development on the technical side was the implementation of our own native tracker software, allowing users to log activities directly within the app rather than relying on a screenshot-based verification system. This advancement marked a shift toward a more integrated solution, enabling a seamless experience for users who could now track their progress within the app itself. Therefore, we added the tracker interface to provide a clearer simulation of the final product, giving participants a more accurate sense of the app's functionality.

Looking ahead, with additional time and resources, this prototype would primarily serve as a foundation to explore further improvements to the tracker's design and user experience. This phase would involve a continued user-centred approach, refining how users interact with the tracking feature and optimising the overall experience based on direct user feedback.

In terms of the user flow, changes were made to account for users being able to get more tasks done without having to change another page. This meant including several ways of logging activity, through the tracker or through the Challenge page. Users could also sort rewards according to recency, company name or type of reward. See Figure 10 below.

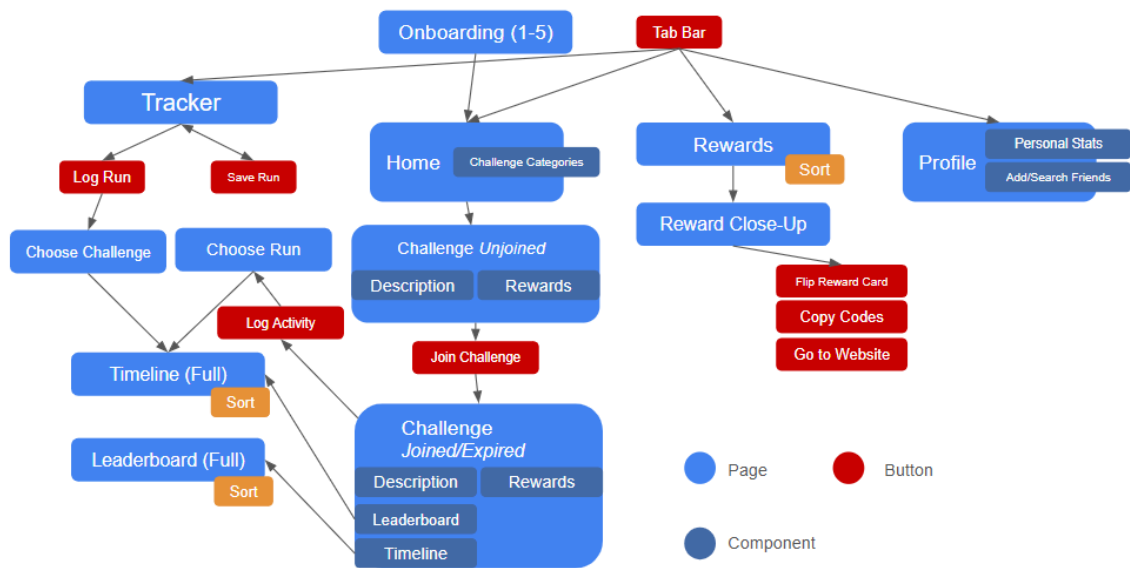


Figure 10: User Flow Diagram of the Final Prototype

6. Technology

The project poses some challenges which will require technical solutions. One major challenge is how to collect user running data like distance, time, date and location. This data will be essential for knowing when a challenge is completed and thus when the user gains access to rewards. Without a system for tracking, our app would be impossible to use, unless we gave out rewards without verifying a run has taken place, which counteracts the goal of motivation - the sense of earning rewards. Similarly, if tracking the runs is unreliable or not user friendly, then using the app will become very frustrating which would discourage users and ultimately decrease our ability to attract and retain customers. To solve this problem, we have identified three main solutions for collecting user data.

6.1 Image Verification

The cheapest and most technically viable solution we came up with was requiring users to upload an image which documents their activity. This image would be a screenshot of a running tracking app of their choice (Strava, runkeeper, etc), where the date, distance and time is provided. The data is then extracted from the image using an optical character recognition library. A very similar solution is done by Relive which allows users to download their activities to a file which is then uploaded to their service (Relive, n.d.). This solution has some advantages, namely that it's quite easy to implement and it allows users to use the tracker of their choice. It also means that we would not actively compete with these companies for users, which would simplify our entry onto the market. Disadvantages with this method is that it requires the user to download an external tracker if they don't already have one as well as it requires the extra steps of taking a screenshot and uploading it. Preferably the used method would require as few extra steps as possible in order to keep the process smooth for the user. Because of the simplicity and viability of the method it was implemented ([Appendix D](#)).

Upon further research, we discovered this method could imply legal issues. The allowance of using another app's data to drive our product varied between different tracking providers, as an example the tracking software Asics Runkeeper terms of service contain the following statement:

"The Services are for personal use only and may not be used in connection with any commercial endeavours except those that are specifically approved by us. The following activities are expressly prohibited: ... (iii) use of web scraping, web harvesting, or web data extraction methods from Runkeeper even if the Runkeeper account owner gives permission" (ASICS Runkeeper, 2023).

Because we have limited knowledge in law and do not have the resources nor contacts to explore this further, and in combination with the previously mentioned disadvantages of the approach, we decided this method of activity verification will likely not be used in the end product.

6.2 Third party API

Another solution is to import the running activities through the APIs of third party run tracking apps such as Runkeeper and Strava. This approach requires a more robust and sophisticated backend but is otherwise quite straightforward since documentation for the API's are readily available. This has the same advantage as the image verification approach, that it allows users to use the running app of their choice and that we would not be active competitors with these providers. Additionally, this approach would mean transferring the run from the run tracking app to our app takes fewer steps for the user, in essence we could either automatically import the users activity or add a simple sync button that would allow users to smoothly import all their activity. From what we have found all major run tracking apps have APIs which means it should be possible to implement a way for users to import their activities regardless of what run tracker they use.

The main issue with this approach is that all different tracking service providers have different terms of service regarding how their API can be used. Some of the providers prohibit the use of their API for commercial use, unless a specific agreement is reached. In our research we found quite a few other services which use third party APIs (such as Zwift, Nike Run Club, Charity Miles, Runtastic, MapMyRun etc.) which shows that it is technically and legally possible, at least with some service providers however, we don't know what kind of agreements these companies has reached. Examining the terms of services for each provider as well as making agreements regarding API usage has been deemed unfeasible for the scope of this project due to our limited legal knowledge and time constraints. Therefore we don't have any estimates of the total cost of using third party APIs in order to obtain our data to power the app. We do however want to allow users to use their favourite trackers in our product, therefore we aim to use this approach rather soon. However, if the legal work required is hard and costly as well as if the agreements are unfavourable, this solution may only be viable long term.

6.3 Native Tracking

Our last proposed way to solve this task is by building our own tracking system which can track distance, time, pace etc. of a user's running session. This use of tracking natively in the app has some advantages and disadvantages. The main advantage is that the user does not need a third party app in order to track their activity, instead the app would be a more holistic solution. This is a great advantage for new runners, since the hurdle to start using the app is smaller if only the app itself and nothing else is needed. On the other hand, integration with third-party devices like smartwatches or other fitness trackers could provide a better experience for users who prefer external devices, or those users who already have a large amount of data logged in other apps and therefore do not wish to

migrate to a different platform. A combination of a native tracker and third party integration would be optimal to satisfy most users' needs.

The tracking system (see [Appendix E](#)) uses React Native and Expo for a consistent iOS and Android environment, see Figure 11. The core logic in the App.tsx file handles tracking sessions, displaying metrics (duration, distance, and pace), and managing user interactions. Session states track activity status, elapsed time, and distance, calculated with a custom hook (useLocationTracking). Another hook, useInterval, updates duration continuously. The App.tsx file also toggles between a map view when inactive and a metrics display when tracking. Further development of this system includes applying filters to location data, to smoothen the route being presented to the user at the end of the run. For the moment, the GPS data contains some noise which makes this data scattered and not as smooth as a user would expect. Additionally, the user interface needs to be tested with users to provide insights into how this technology truly can enhance the user experience and provide additional value that our competitors do not.

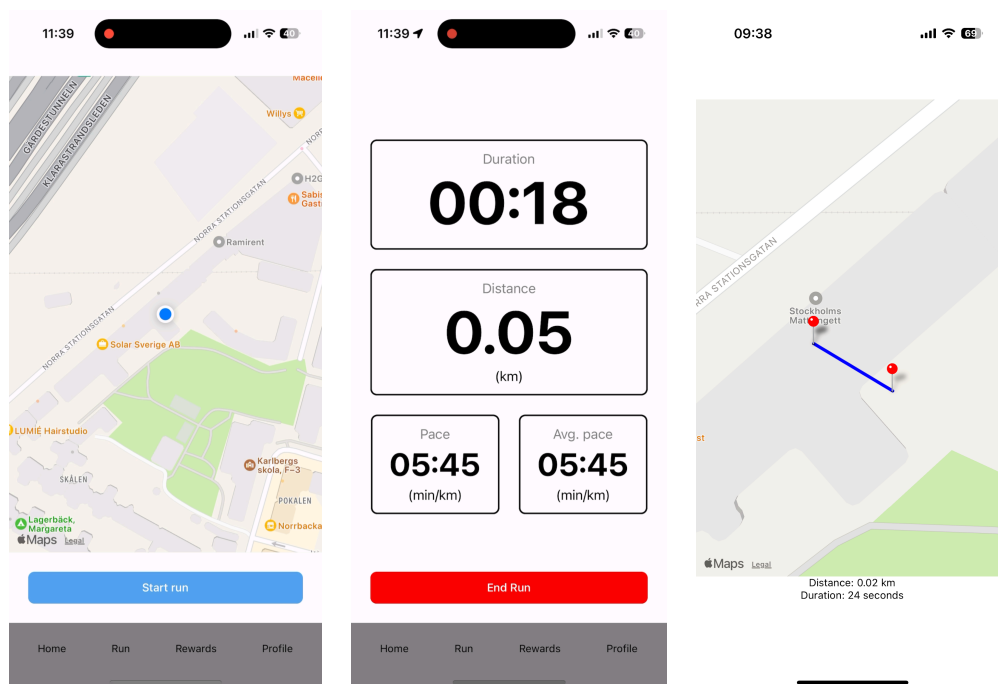


Figure 11: The tracking software in action

The configuration files set essential parameters. The app.json file, specific to Expo, holds metadata such as app name, orientation, and icon settings, with platform-specific adaptations for Android and iOS. babel.config.js enables JSX and TypeScript, enhancing compatibility and performance, while package.json includes libraries for GPS tracking, map visualisation, and core dependencies.

Additional components (PrimaryButton, Map, MetricBox, and NavBar) modularize functionality: PrimaryButton controls session start/stop, Map dynamically displays the user's location, and MetricBox shows session metrics. Each component includes TypeScript definitions for structured use. Finally, icons and splash screens in the assets directory, referenced in app.json, ensure consistent branding and a professional experience across devices.

6.4 Summary Technical Solutions

Our investigation has shown that the image verification approach is rather messy and therefore this idea has been scrapped. Both the third party API and native tracking did however provide quite a few

advantages as well as some challenges. Going forward we aim to allow for both tracking using third party apps and tracking natively in our app. This provides the major advantages that it doesn't introduce a hurdle for new users to download a second app and integrate the two, as well as it doesn't force users which already use a specific tracking app to switch to our system. Combining the two solutions offers the best of both worlds and minimises the hurdles for all types of users. The difficulties are instead passed on to us, the developers. Using both solutions requires both implementing our own tracking system and getting familiar with the terms of services of different API providers as well as potentially working with different agreements. The first challenge seems quite doable based on how easy we could get a prototype running. The second challenge seems to be very technically easy, however, it will require some legal work which we are all very unfamiliar with and therefore it may require external assistance.

7. Conclusion

Our PricePal app aims to address societal issues related to sedentary behaviour and limited access to affordable fitness activities, especially in an era where the rising cost of living impacts participation in structured physical activities. By using financial incentives and fitness challenges, the app motivates users to stay active while providing value through partnerships with brands and companies. This concept aligns with existing research, which emphasises the effectiveness of rewards in encouraging consistent physical activity and habit formation.

The design was developed throughout the whole project, starting from the initial brainstorming session through various iterations of design and user testing. We have explored potential challenges, including legal and technical hurdles, and proposed viable solutions for integrating third-party APIs and native tracking features. From a business standpoint, the app has a unique niche by combining elements of community, competition, and brand engagement. It sets itself apart from competitors like Mecenat and Stepler by promoting deeper brand interaction through effort-based rewards, which is strengthened by psychological principles such as the IKEA effect. Also, our market survey indicated that advertisers see potential in a model that requires user participation to earn rewards, although challenges remain in proving the added value compared to higher-volume platforms. The financial analysis demonstrates a clear path to profitability, with a positive Net Present Value (NPV) highlighting the project's long-term viability.

7.1 Internal goals

We set several internal goals for this project, which we achieved either fully or to varying degrees. Our first goal was to create a high-fidelity prototype that we would be proud to showcase in our portfolios. We believe we achieved this goal, as we are generally satisfied with the design of our prototype, which was also confirmed in our second round of user testing.

Our second goal was to ensure that the idea was clear enough for (potential) users and advertisers to understand our value proposition. Feedback from both user testing and our market survey indicated that users understood the concept of the app and that advertisers, in several cases, found our value proposition more attractive than that of our competitors, as they appreciated our focus on user engagement. However, it was unclear whether advertisers fully grasped our reasoning behind the IKEA effect – the idea that the effort put in by the user increases their perceived value of the product. We acknowledge that this concept is not immediately obvious or easy to communicate. Overall, we believe we achieved this goal.

Our third goal was to implement a functioning validation system. Our initial approach was to use image validation, for which we developed a working solution. However, we later decided that this validation system would not be used and began exploring the implementation of third-party API imports. We found that this approach required a deeper understanding of providers' terms of use and potentially required contracts, so it was not implemented. Finally, we worked on a prototype of our own tracking system, though this prototype lacks certain features we would have liked and remains low fidelity. Based on this, we believe we have partially reached this goal, as we developed a prototype and a pathway toward a finished solution, though not a complete implementation.

Last but not least, our fourth goal was to utilise our diversity as a cross-functional team. We worked in a way that allowed us to learn from each other and enabled each team member to contribute their unique skills and strengths. Although everyone participated in most processes, the design-focused team members contributed more to the design and user testing, while the management-focused members provided more insights into the business perspective, analyses, and surveys. The technical perspective was shared fairly equally, as all team members had the appropriate abilities. Beyond using our skills to advance the project, we all enjoyed the collaborative experience a lot and gained valuable knowledge by working with people we might not otherwise have met. Based on this, we believe we achieved this goal fully.

7.2 Future Work

To build on the foundation laid in this project, we have identified several areas for future development and improvement. First, further exploration of third-party API agreements will be essential to ensure long-term sustainability and compliance with legal standards. Engaging legal experts to help navigate terms of service and establish favourable agreements with popular running app providers like Strava and Runkeeper will be crucial for this step. Additionally, enhancing the app's native tracking capabilities to include integrations with wearable technology such as smartwatches and fitness bands will expand the user base and strengthen the tracking experience.

Secondly, user experience improvements should be a continuous focus. Future iterations should include refined user feedback mechanisms during pilot studies and subsequent rounds of user testing to gather more data on app usability and user satisfaction. Expanding the scope of rewards beyond running-related products to include a wider range of offers from various industries will help maintain user interest and engagement over time. Marketing efforts should also explore partnerships with non-fitness-related brands that share values aligned with health, wellness, and sustainable living.

Finally, on the technical side, optimising data handling for real-time tracking and improving the accuracy of distance and pace calculations are areas for refinement. This will ensure a seamless user experience and foster trust in the app's reliability. Additionally, developing more sophisticated algorithms for personalised challenges and adaptive difficulty levels could help keep users motivated and engaged, encouraging continued use of the platform.

In conclusion, while the initial development phase has demonstrated the viability and potential of the PricePal app, future work will focus on expanding its capabilities, refining the user experience, and ensuring sustainable growth through strategic partnerships and technical enhancements. Addressing these areas will hopefully position the app as a leading tool in promoting physical activity, community engagement, and brand interaction.

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Appendix

Appendix A

The market survey can be accessed through the following link: <https://forms.gle/rz1MgVeCUVbGaDBb6>

Appendix B

The answers to the Market Survey can be accessed through the following link:

<https://docs.google.com/spreadsheets/d/1RMDfq7efp3SQvpbenYY9ddrAOFRsKNNwg-L7eOlik2g/edit?usp=sharing>

Appendix C

The Figma Prototype can be accessed through the following link:

<https://www.figma.com/design/xCqT2aGAGepzMFrBeIRUFs/KID%2FDEL-Prototype?node-id=0-1&t=Fj4YCZTSAb2MmbGr-1>

Appendix D

The image verification implementation for a runkeeper screenshot can be found on GitHub using the following link: <https://gits-15.sys.kth.se/antdre/KID-DEL-ImageRec.git>

Appendix E

The implementation of the native tracking system can be accessed through GitHub using the following link:

<https://github.com/larssonpatrik/KID-RunningTracker.git>